Free Resolutions

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This is a note discussing the basic construction of Free Resolutions and the computation of free resolutions for various modules.

1 Basic Construction

Let R be a commutative ring with 1 and let M be a finitely generated module over R. If M is generated by $x_1, \cdots, x_n$ elements with m relations f_i between the generators then recall that we can encode the presentation of M with an exact sequence

$$R^m \to R^n \to M \to 0$$

where $R^n \to M$ is given by $a_i \to x_i$ and where $R^m \to R^n$ is given by $b_i \to f_i(a_i)$. Having a presentation for a module is already pretty good, but we can go further. Recalling that R-mod is an abelian category gives us that $\ker(R^m \to R^n)$ is also an R-module. And so we can ask for a presentation of $\ker(R^m \to R^n)$. This gives us a further exact sequence

$$R^k \to R^m \to R^n \to M \to 0.$$

We can then iterate this procedure. This process may not terminate (even when M is fg? we should find an example). Repeating this procedure gives the *free resolution of M*

$$\cdots \to R^{k_n} \to \cdots R^{k_2} \to R^{k_1} \to R^{k_0} \to M \to 0.$$

This is called "Free" since every object in the chain (except the final one) is a free module.

Practically speaking, here's another way to think about the iteration of this process which may help with the computations. M is finitely generated with n_1 generators and so we start with the exact sequence

$$R^{n_1} \to M \to 0.$$

Now we ask for the generators of $\ker(R^{n_1} \to M)$; say this has n_2 generators. We can then extend our exact sequence

$$R^{n_2} \to R^{n_1} \to M \to 0$$

Now, ask for the generators of $\ker(R^{n_2} \to R^{n_1})$,

$$R^{n_3} \to R^{n_2} \to R^{n_1} \to M \to 0.$$

Then continue this process indefinitely, or until $\ker(R^{n_{k+1}} \to R^{n_k}) = 0$ is the zero module[1]. Need to double check that this is equivalent to the first construction given.

Below we will compute some examples of free resolutions of various modules.

2 Examples

These examples are sorted into an order of increasing complexity. However, in all the following examples, R is a polynomial ring and M is a quotient of R. One can check these examples in Macaulay2 with the code what's the use of verbatim

Example 1. Let $R = k[x]$ and $M = R/(x^n)$ for some fixed n. Then notice that M is generated by $\bar{1} \in R/(x^n)$ as an R-module. Moreover, we only have a single relation $x^n \cdot \bar{1} = 0$. We can then capture this in the presentation

$$R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto \bar{1}} M \to 0.$$

The kernel of $R^1 \to M$ is exactly the ideal (x^n). Interpreting (x^n) as an R-module we find that it is a rank 1 free module (generated by x^n). That is a presentation for (x^n) as an R-module is

$$0 \to R^1 \xrightarrow{1 \mapsto x^n} (x^n).$$

Now since $\ker(R^1 \to (x^n)) = 0$ we can no longer iterate our procedure. Alternatively $\ker(R^1 \xrightarrow{1 \mapsto x^n} R^1) = 0$. This leaves us with the free resolution of $M = R/(x^n)$.

$$R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto \bar{1}} M \to 0.$$

This example shows us that whenever our kernel module is free then we are done.

[1]Note that at each stage of the above construction we are taking the kernels to be finitely generated. This is only necessarily true if R is Noetherian.

Example 2. Let $R = k[x,y]$ and $M = R/(x^2,y^2)$, elements in M are of the form $f = a_0 + a_1 x + a_2 y + a_3 xy$. As an R-module, M is generated by $\bar{1} \in M$. We have the relations $x^2 = 0$ and $y^2 = 0$, iin fact, $\ker(R^1 \to M)$ is exactly the ideal (x^2, y^2). A presentation for M is thus

$$R^2 \xrightarrow{\begin{bmatrix} x^2 & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \to 0$$

Now, the kernel module $\ker(R^1 \to M) = (x^2, y^2)$. This module is generated by two elements x^2, y^2 and we have the R-linear relation that $y^2(x^2) - x^2(y^2) = 0$. Thus[2]our next iterated sequence is

$$R^1 \xrightarrow{\begin{bmatrix} y^2 \\ -x^2 \end{bmatrix}} R^2 \xrightarrow{\begin{bmatrix} x^2 & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \to 0.$$

Now notice that the kernel module $\ker(R^1 \to R^2) = 0$ is the zero module, and so we are done. The above sequence is thus the free resolution for $M = R/(x^2, y^2)$.

Example 3. Let $R = k[x,y]$ and $M = R/(x,y)^2$. Note $(x,y)^2 = (x^2, xy, y^2)$. Now M is generated as an R-module by $\bar{1} \in M$ but we have the relations $x^2 = 0, xy = 0, y^2 = 0$. This gives the presentation

$$R^3 \xrightarrow{\begin{bmatrix} x^2 & -xy & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \to 0.$$

We next want to understand the kernel module $\ker(R^3 \to R^1)$. If $(a,b,c) \in R^3$ then this triple is in the kernel module when $ax^2 + bxy + cy^2 = 0$. Notice that the two vectors $(y, -x, 0)$ and $(0, y, -x)$ are both vectors[3] in the kernel module. We claim that these are in fact an R-basis of $\ker(R^3 \to R^1)$. The argument we have is a direct computation and is a little tedious. For clarity we will put the details in the appendix and continue the remainder of the argument for computing the free resolution.

We have that $\ker(R^3 \to R^1)$ is spanned by the vectors $(y, -x, 0), (0, y, -x)$ and so we extend our exact sequence

$$R^2 \xrightarrow{\begin{bmatrix} y & 0 \\ -x & y \\ 0 & -x \end{bmatrix}} R^3 \xrightarrow{\begin{bmatrix} x^2 & -xy & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \to 0.$$

Now, continuing the process, we want to understand $\ker(R^2 \to R^3)$ where $R^2 \to R^3$ is the map

[2] To be rigorous, one needs to argue that for any $(a,b) \in R^2$ such that $ax^2 + by^2 = 0$ that we can write (a,b) as an R-linear combination of (y^2, x^2). We will do an example of this calculation in the next example.

[3] These are called "syzigies" and are understood as relations amoung the relations.

$(a, b) \mapsto (ay, -ax + by, -bx)$. That is, we want to understand for which $[a, b]^T \in R^2$ we have

$$\begin{bmatrix} ay \\ -ax + by \\ -bx \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

However, since R is a domain, we have that this is only satisfied by $a = b = 0$. Thus, $\ker(R^2 \to R^3) = 0$ and we are done. Our free resolution of $M = R/(x, y)^2$ is thus

$$R^2 \xrightarrow{\begin{bmatrix} y & 0 \\ -x & y \\ 0 & -x \end{bmatrix}} R^3 \xrightarrow{\begin{bmatrix} x^2 & xy & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \to 0.$$

Appendix Calculation. Here we provide the details of the argument that $\ker(R^3 \to R^1) = \langle (y, -x, 0), (0, y, -x) \rangle$. Suppose $(a, b, c) \in R^3$ such that $ax^2 + bxy + cy^2 = 0$. For this relation to hold true we must have $y|a$, otherwise the term ax^2 could not be cancelled by the other terms. A similar argument gives that we must have $x|c$. Write $a = fy$ and $c = gx$ for $f, g \in R$.

Let $\phi : R^3 \to R^1$ denote our R-linear map; i.e. $\phi(j, k, l) := jx^2 + kxy + ly^2$. Consider the vector

$$\begin{aligned} \underline{x} &= \begin{bmatrix} a \\ b \\ c \end{bmatrix} - f \begin{bmatrix} y \\ -x \\ 0 \end{bmatrix} - g \begin{bmatrix} 0 \\ y \\ -x \end{bmatrix} \\ &= \begin{bmatrix} fy \\ b \\ gx \end{bmatrix} - f \begin{bmatrix} y \\ -x \\ 0 \end{bmatrix} - g \begin{bmatrix} 0 \\ y \\ -x \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ b + fx - gy \\ 0 \end{bmatrix}. \end{aligned}$$

Now, by R-linearity of ϕ we have that $\phi(\underline{x}) = 0$, in otherwords $xy(b + fx - gy) = 0$. Now, since R is a domain, we must actually have that $b + fx + gy = 0$ is a relation. Thus $\underline{x}$ is actually the 0 vector in R^3. That is, given $(a, b, c) \in \ker(R^3 \to R^1)$ we have shown that there exists $f, g \in R$ such that $(a, b, c) = f(y, -x, 0) + g(0, y, -x)$. That is $(y, -x, 0), (0, y, -x)$ span $\ker(R^3 \to R^1)$. Moreover, since R is an integral domain, we have that $(y, -x, 0), (0, y, -x)$ are also R-linearly independent, and so in fact, $(y, -x, 0), (0, y, -x)$ is a basis for $\ker(R^3 \to R^1)$.